

of the article will demonstrate two examples: a blinking LED circuit and a circuit to read temperature accurately on a digital multimeter based on MSP430F2013 chip. The pin details of a 14-pin MSP430F2013 device are shown in Fig. 3.

MSP430F2013 features a 16MHz CPU, 2kB Flash and 128-byte RAM. It has a 16-bit sigma-delta ADC, about ten ports with interrupt capability and in-built pull-up/pull-down resistors, 16-bit versatile timer with two outputs, 16-bit watchdog timer, SPI and I²C interfaces, and a zero-power-brownout

PARTS LIST

Semiconductors:

IC1	- 74AH244N octal tristate buffer/driver
IC2	- MSP430F2013N microcontroller
TL1, TL2	- BC547 npn transistor
ZD1, ZD2	- 3.6V zener diode
SD1, SD2	- 1N5817 schottky diode
PR1	- TL431 precision shunt regulator
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, 5% carbon):

R1	- 1.5-kilo-ohm
R2, R3, R8	- 100-ohm
R4-R7, R9, R10, R14, R20-R23	- 33-kilo-ohm
R11, R12	- 330-kilo-ohm
R13, R17-R19, R21	- 330-ohm
R15	- 6.8-kilo-ohm
R16, R24	- 3.3-kilo-ohm
R25	- 47-kilo-ohm
R26, R28	- 10-kilo-ohm
R27	- 680-kilo-ohm
R29	- 1-kilo-ohm
TM	- 82-kilo-ohm thermistor

Capacitors:

C1, C4, C6	- 0.1µF ceramic
C2, C5	- 10µF, 10V electrolytic
C3	- 2.2nF ceramic
C7	- 470nF ceramic

Miscellaneous:

CON1	- 25-pin, D-type parallel-port female connector
CON2	- USB type B connector
CON3, CON4	- 14-pin JTAG/FRC male connector
SI	- Tactile switch
DMM	- Digital multimeter
BATT	- 3V lithium cell
	- Cell holder
	- 14-pin FRC cable with female connector
	- 25-pin FRC cable with male connector
	- Type-A USB cable

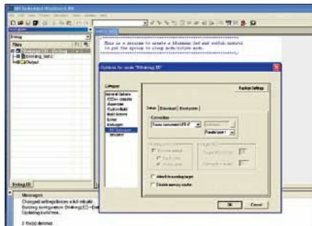


Fig. 4: Screenshot of a typical IAR embedded workbench environment

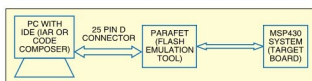


Fig. 5: Block diagram of MSP430 development system

protection block. Brownout protection ensures system safety during power-on or power-off. One of the main reasons for selecting this device is its availability in DIP package, which makes it easy for hobbyists to handle.

Tools for programming and debugging

Every MSP430 chip has a JTAG interface that is used for downloading the user program into the Flash memory of the microcontroller and running and debugging the code. For this, the following tools are required:

1. Development software that resides in the PC. These are typically called integrated development environments (IDEs). Following choices are available:

(a) **IAR Embedded Workbench Kickstart version.** This freeware from IAR Systems is sufficient for developing simple programs in 'C' with size limit of 4 kB. However, there is no such limitation for code development in Assembly language. A screenshot of a typical IAR embedded workbench environment is shown in Fig. 4.

(b) **Code Composer Essentials Evaluation.** This freeware from Texas In-

struments is based on industry-standard Eclipse open-source platform. You can use it to develop code up to 8kB size.

Either of the above software can be downloaded from Texas Instruments' website by selecting 'tools & software' option under the MSP430 ultra-low-power microcontrollers section.

2. Hardware tool called the Flash emulation tool (FET). It connects the PC to the MSP430-based system. Two types of FET tools are avail-

able: One is based on the parallel port, while the other is based on the USB interface.

Fig. 5 shows the block diagram of the MSP430 development system. The next section describes the construction of Flash Emulation Tool based on PC's parallel port called ParafET, which can be used to interface with any of the MSP430 target devices.

Designing MSP430 Flash emulation tool

Fig. 6 shows the ParafET circuit or Flash emulation tool based on the PC's parallel port. As you can notice, the PC connects to this board through a 25-pin, D-type connector. The JTAG interface is established through the 14-pin FRC connector. It is important to note that the signals on the PC side are TTL-compatible and referenced to 5V, while the MSP430 works off 1.8V-3.6V supply. For Flash programming, the voltage needs to be a minimum of 2.2V.

The main component that is used to interface the signals from the PC side to the MSP430 is the octal tristate buffer based on SN74AH244 or SN-74HC244 IC. This chip along with the